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Ans 1. ~~Newton's first law~~ ^{second law}: rate of change of momentum proportional to applied force, direction of force. $F=ma$.

Ans 2. Speed - scalar, magnitude only ~~no direction needed here~~. Velocity - vector, has direction. eg 40km/hr ^{north}

Ans 4. SI unit of ~~voltage~~ ^{of current} is Ampere. $1A = 1 \text{ coulomb/sec}$.

Ans 6. $u=0$ $a=2$ $t=5$

$$v=u+at = 0+2 \times 5 = 10 \text{ m/s}$$

$$s=ut + \frac{1}{2}at^2 = 25\text{m} \text{ (smudged)}$$

Ans 8. $V=IR$ ~~$V=20$~~ $V=2 \times 10 = 20 \text{ Volts}$ (scratched out mistake above)

Ans 11 (a). $v=u+at$, avg vel $= (u+v)/2$, $s = \text{avg vel} \times t$ ^{substituting v}: $s = ut + \frac{1}{2}at^2$
 messy margin note:
 "check this!"

Ans 9. 3 states: ~~solid, gas, liquid~~ ^ solid, liquid, gas – particle spacing differs.

Ans 7. Mitosis=2 identical cells(growth). Meiosis=4 cells ~~same~~ ^ half chromosomes(reproduction).

Ans 12. 1)rainwater harvesting 2)fixing leaks [coffee stain smudge]

rough work / formulas (ignore): $F=ma$ $v=u+at$ $s=ut+.5at^2$